

Unbounded normalized cofactor spectra for rank two Gram matrices

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Abstract

For a positive-semidefinite Hermitian matrix A , let $C(A)_{ij} = a_{ij} \operatorname{per} A(i | j)$, where the minor deletes row i and column j . We prove that $\lambda_{\max}(C(A))/\operatorname{per} A$ has no dimension-independent upper bound, already for complex rank-two correlation matrices. The same construction makes $\lambda_{\max}(\operatorname{Re} C(A))/\operatorname{per} A$ arbitrarily large. The proof computes a spectral limit under repeated Gram vectors and combines it with projective equidistribution. It also gives positive-definite examples and unbounded local second-order relative coefficients for entrywise-positive real correlation perturbations.

1 Statement and scope

Our convention is that $A(i | j)$ deletes row i and column j , without transposing this minor. All Hermitian inner products below are linear in their first argument. The matrix $C(A)$ is positive semidefinite and $C(A)\mathbf{1} = (\operatorname{per} A)\mathbf{1}$. Drury disproved the assertion that this prescribed eigenvalue is always the largest one [1]. Here we consider all constant multiples of the permanent.

Theorem 1. *For every $R > 0$ there are an integer N and a complex Hermitian correlation matrix A of rank exactly two such that*

$$\lambda_{\max}(C(A)) > R \operatorname{per} A, \quad \lambda_{\max}(\operatorname{Re} C(A)) > R \operatorname{per} A.$$

There are also positive-definite correlation matrices with both inequalities. The positive-definite assertion does not retain rank two.

The real part is entrywise, so its largest eigenvalue is the maximum Rayleigh quotient over real unit vectors. The theorem is an unboundedness assertion as the order varies; no rate in N is asserted. It is distinct from unbounded Hadamard-product bunching ratios [3], and from bounds for the full Schur-power matrix. It does not resolve Lieb’s normalized character or subgroup permenental-dominance conjecture [4]. A separate source audit records the finite literature search; no claim of exhaustive priority verification is made.

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2 A repetition limit

Let $v_i = (a_i, b_i)$ be nonzero unit rows, $1 \leq i \leq n$, put $\ell_i(x, y) = a_i x + b_i y$, and set $P = \prod_i \ell_i$ and $A_{ij} = v_i v_j^*$. For binary forms of degree d define

$$\left\langle \sum_{k=0}^d p_k x^{d-k} y^k, \sum_{k=0}^d q_k x^{d-k} y^k \right\rangle_B = \sum_{k=0}^d (d-k)! k! p_k \overline{q_k}.$$

Leibniz expansion gives

$$\left\langle \prod_i \ell_i, \prod_j \ell'_j \right\rangle_B = \text{per}[\langle \ell_i, \ell'_j \rangle_B]. \quad (1)$$

In particular $\text{per } A = \|P\|_B^2 > 0$, and $C(A)$ is the Gram matrix of $v_i \otimes (P/\ell_i)$. This proves its positivity with the stated conventions. For Haar probability measure μ on $\mathbb{CP}^1$, represented by unit vectors,

$$\langle p, q \rangle_B = (d+1)! \int p \overline{q} d\mu. \quad (2)$$

Indeed, phase integration makes distinct monomials orthogonal and $\int |x|^{2a} |y|^{2b} d\mu = a!b!/(a+b+1)!$.

Repeat each row L times, write $N = nL$, and let A_L be the resulting Gram matrix. Then $P_L = P^L$ and, with cluster indices (i, a) ,

$$C(A_L)_{(i,a),(j,b)} = A_{ij} \langle P^L/\ell_i, P^L/\ell_j \rangle_B. \quad (3)$$

Let $U_L : \mathbb{C}^n \rightarrow \mathbb{C}^{nL}$ map e_i to the vector equal to $L^{-1/2}$ on cluster i and zero elsewhere. All clusters in (3) are constant, so $C(A_L)$ vanishes on $\text{ran}(U_L)^\perp$. Its nonzero normalized spectrum, including multiplicities, is that of $K_L = U_L^* C(A_L) U_L / \text{per } A_L$. Writing $d\mu_L = |P|^{2L} d\mu / \int |P|^{2L} d\mu$, the factorials in (2) give the exact formula

$$(K_L)_{ij} = \frac{L}{nL+1} A_{ij} \int \frac{d\mu_L}{\ell_i \ell_j}. \quad (4)$$

The expression against the density at a zero is interpreted through the polynomial $(P^L/\ell_i)(\overline{P^L/\ell_j})$.

Lemma 2. *Suppose $|P|$ has a unique projective maximum. Rotate it to $(1, 0)$ and put $z_i = b_i/a_i$. Then every $a_i \neq 0$, $\sum_i z_i = 0$, and*

$$K_L \longrightarrow \frac{J_n + z z^*}{n} \quad \text{in operator norm.}$$

Consequently

$$\lim_{L \rightarrow \infty} \frac{\lambda_{\max}(C(A_L))}{\text{per } A_L} = \max \left(1, \frac{\|z\|^2}{n} \right), \quad (5)$$

$$\lim_{L \rightarrow \infty} \frac{\lambda_{\max}(\text{Re } C(A_L))}{\text{per } A_L} \geq \max \left(1, \frac{\|z\|^2}{2n} \right). \quad (6)$$

Proof. The maximum of the nonzero polynomial is positive, so no factor vanishes there. The integrand in (4) is continuous near the maximum u . Outside any neighborhood of u , $|P| \leq a < |P(u)|$. On a smaller neighborhood of positive measure, $|P| \geq b > a$. The absolute unnormalized tail is bounded by

$$a^{2L-2} \sup |(P/\ell_i)(P/\ell_j)|,$$

whereas the denominator is bounded below by a positive constant times b^{2L} . Thus the apparent poles cause no problem and the integral tends to $(a_i \overline{a_j})^{-1}$. Since n is fixed, entrywise convergence implies operator-norm convergence and gives the stated matrix limit.

In the local coordinate $(1, t)/\sqrt{1 + |t|^2}$,

$$\log |P|^2 = \text{constant} + \sum_i \log |1 + z_i t|^2 - n \log(1 + |t|^2).$$

The two real first derivatives at zero vanish, hence $\sum_i z_i = 0$. The limiting matrix therefore has eigenvalues 1, $\|z\|^2/n$, and zeros, which proves (5). For $z = x + iy$ its real part is $(J_n + xx^T + yy^T)/n$, with $x, y \perp \mathbf{1}$. The last two terms have rank at most two and trace $\|z\|^2/n$, proving (6). Compression commutes with real part because U_L is real. $\square$

3 Equidistribution and selection of one maximum

Choose collections of m unit rows whose empirical projective measures converge weakly to Haar measure on $\mathbb{CP}^1$. Such a sequence is obtained from midpoint grids in $p = |a|^2 \in (0, 1)$ and the relative phase, with both grid sizes tending to infinity. Let u_m be any maximum of the corresponding product, and rotate variables so that $u_m = (1, 0)$.

These moving rotations preserve weak convergence to Haar. Indeed, every subsequence of rotations has a convergent subsubsequence by compactness of $U(2)$. For each continuous test function, the rotated functions then converge uniformly; weak convergence and Haar invariance give the same limiting integral. This subsequence argument proves the claim for the full sequence.

For $T > 0$ the function

$$f_T([a : b]) = \min(|b/a|^2, T), \quad f_T([0 : 1]) = T,$$

is continuous. Since $|a|^2$ is uniform on $[0, 1]$ under Haar measure,

$$\int f_T d\mu = \int_0^1 \min\left(\frac{1-p}{p}, T\right) dp = \log(1 + T). \quad (7)$$

Every a_i in the rotated configuration is nonzero. It follows that

$$\liminf_{m \rightarrow \infty} \frac{1}{m} \sum_{i=1}^m |b_i/a_i|^2 \geq \log(1 + T) \quad \text{for every } T > 0.$$

Hence these averages tend to infinity. Only bounded continuous test functions have been used; there is no assumption of convergence of an unbounded moment.

To make the maximum unique, append one unit linear form $\ell_u(z) = \langle z, u_m \rangle$, linear in z . On the unit sphere its modulus is at most one, with equality only at the projective point u_m . Multiplication by this factor leaves the old maximum value unchanged and makes its location unique. In the rotated coordinates the added row is $(1, 0)$, contributing $z = 0$. For the augmented base $n = m + 1$ we still have

$$\frac{1}{n} \sum_{i=1}^n |z_i|^2 = \frac{1}{m+1} \sum_{i=1}^m |b_i/a_i|^2 \rightarrow \infty. \quad (8)$$

Proof of Theorem 1. Given R , first choose one finite base in (8) with average exceeding $2R + 2$. Equidistribution ensures that a sufficiently large base spans $\mathbb{C}^2$. Freeze it and then take a sufficiently large integer L in Lemma 2. Both ratios exceed R , all rows are unit, and the Gram matrix has rank exactly two.

For a strict witness A , polynomial dependence of $C(A)$ and per A , continuity of Hermitian eigenvalues, and per $A > 0$ show that the strict inequalities persist for $A + \varepsilon I$ with sufficiently small $\varepsilon > 0$. The matrix $(A + \varepsilon I)/(1 + \varepsilon)$ is a positive-definite correlation matrix. Both ratios are unchanged by this final scalar rescaling, since $C(cA) = c^N C(A)$ and $\text{per}(cA) = c^N \text{per } A$. $\square$

4 A local consequence and an exact finite certificate

The following is the real-direction specialization of the perturbation formula of Pioge and coauthors [2, Lemma 1]; we include the calculation to fix the Gram and conjugation conventions. For a real unit vector w , let

$$B(\varepsilon)_{ij} = \frac{1 + \varepsilon^2 w_i w_j}{\sqrt{(1 + \varepsilon^2 w_i^2)(1 + \varepsilon^2 w_j^2)}}.$$

This is a real correlation matrix of rank at most two, with strictly positive entries for sufficiently small ε . Expanding the entries and then the permanent gives

$$\frac{\text{per}(A \circ B(\varepsilon))}{\text{per } A} = 1 + \varepsilon^2 \left(\frac{w^T \text{Re } C(A) w}{\text{per } A} - 1 \right) + O(\varepsilon^4). \quad (9)$$

In fact the quadratic coefficient before division by per A is

$$\sum_{ij} C(A)_{ij} \left(w_i w_j - \frac{w_i^2 + w_j^2}{2} \right) = w^T C(A) w - \text{per } A = w^T \text{Re } C(A) w - \text{per } A,$$

using both row and column sums. Theorem 1 therefore makes the relative coefficient in (9) arbitrarily large. This is a local statement; it gives no uniform finite- ε gain or perturbative interval. For complex w , the column-Gram convention in [2] has numerator $1 + \varepsilon^2 \overline{w_i} w_j$; replacing it by the opposite convention would conjugate the Rayleigh direction.

For an optional finite check, the companion file `witness_n200.csv` specifies 200 Gaussian-integer rows $v_i \in \mathbb{Z}[i]^2$ and a vector $w \in \mathbb{Z}[i]^{200}$. The standard-library integer verifier proves

$$\frac{269}{100} < \frac{w^* C(VV^*) w}{\text{per}(VV^*) \|w\|^2} < \frac{27}{10}.$$

It independently forms each omitted-factor polynomial and evaluates $\sum_{a=1}^2 \|\sum_i \overline{w_i} v_{ia} (P/\ell_i)\|_B^2$. This certificate illustrates a finite violation and checks the conventions; unboundedness follows from the analytic proof, not from numerical extrapolation.

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```
python3 verify_witness.py
python3 verify_witness_independently.py
```

The computational checks do not replace the analytic proof.

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